ROS 2 Heading Output Guide

Live gyro, magnetometer, and Kalman-filtered headings

What this guide runs

The Kalman-filtered heading is calculated by heading_live and published on /imu/heading_kalman (std_msgs/msg/Float64, radians). print_headings subscribes to it, prints it in the terminal, and plots it with the gyro and magnetometer headings.

Before starting

- Use a ROS 2 Linux / WSL terminal with the package in a colcon workspace.
- Connect the IMU and make sure its configured I2C bus is available.
- Have both calibration files ready in ~/imu_calibration/:

```
bias_file_gyro.csv
```

```
mag_calibration.csv
```

If calibration files are missing, first follow the IMU calibration instructions in the project README. Keep the rover stationary while recording gyro bias.

Build the package once

Open a terminal at the colcon workspace root (the folder containing src/). Replace <ros_distro> with your installed ROS 2 distribution, such as humble or jazzy.

```
source /opt/ros/<ros_distro>/setup.bash
```

```
colcon build --packages-select imu_processing
```

```
source install/setup.bash
```

Open four terminals for the four nodes below. In each new terminal, source ROS and the workspace overlay before running its command:

```
source /opt/ros/<ros_distro>/setup.bash
```

```
cd <workspace_root>
```

```
source install/setup.bash
```

Run the live heading pipeline

Terminal 1 - raw IMU publisher

```
ros2 run imu_processing imu_raw_live
```

This reads the ICM-20948 and publishes /imu/gyr_raw and /imu/mag_raw. Keep this terminal running.

Terminal 2 - calibration node

```
ros2 run imu_processing imu_cal_live
```

This subscribes to the raw topics and publishes /imu/gyr_cal and /imu/mag_cal using the two calibration files.

Terminal 3 - heading and Kalman filter

```
ros2 run imu_processing heading_live
```

This publishes /imu/heading/gyro, /imu/heading/mag, and the fused /imu/heading_kalman.

Terminal 4 - print and plot

```
ros2 run imu_processing print_headings
```

The terminal prints the latest headings about five times per second. The plot is enabled by default and shows radians in the range [-pi, pi]. Keep this process running to see live updates.

Expected terminal output

```
gyro_head: 0.123456, mag_head: 0.130000, kalman_head: 0.127891
```

Values are in radians. At startup, a field may show waiting until its first message arrives. The Kalman estimate starts from the first magnetometer heading, then combines gyro prediction with magnetometer corrections.

Check topics if there is no output

```
ros2 topic list | grep /imu
```

```
ros2 topic echo /imu/heading_kalman
```

Confirm all four nodes are running in terminals on the same ROS_DOMAIN_ID and that each terminal sourced the same ROS distribution and workspace.

No graphical display available?

```
ros2 run imu_processing print_headings --ros-args -p enable_plot:=false
```

This disables the matplotlib window but keeps terminal output. If a node cannot find a calibration file, check ~/imu_calibration and the calibration_dir parameter.

The magnetometer heading uses horizontal-plane atan2 and is not tilt compensated. For accurate heading, keep the rover level and verify sensor-axis alignment and calibration.